1 **(a)** $\frac{45}{360} \times 2 \times \pi \times 12$ **[M1]**

3π **[A1]**

(b) $\frac{45}{360} \times \pi \times 12^2$ **[M1]**

18π **[A1]**

[Total Marks: 4]

2 **(a)** $\frac{60}{360} \times 2 \times \pi \times 12$ **[M1]**

4π **[A1]**

(b) $\frac{60}{360} \times \pi \times 12^2$ **[M1]**

24π **[A1]**

[Total Marks: 4]

3 **(a)** $\frac{45}{360} \times \pi \times 12^2$ **[M1]**

18π **[A1]**

(b) $\frac{45}{360} \times 2 \times \pi \times 12$ **[M1]**

3π **[A1]**

$3\pi + 24$ **[A1]**

[Total Marks: 5]

4 **(a)** $\frac{150}{360} \times \pi \times 6^2$ **[M1]**

15π **[A1]**

(b) $\frac{150}{360} \times 2 \times \pi \times 6$ **[M1]**

5π **[A1]**

$5\pi + 12$ **[A1]**

[Total Marks: 5]

5 **(a)** $\frac{\theta}{360} \times \pi r^2 = 10\pi$ **[M1]**

$\frac{\theta}{360} \times 2\pi r = 2\pi$ **[M1]**

Valid method to solve simultaneously (e.g. dividing the area equation by the arc length equation to get $\frac{r}{2} = 5$) **[M1]**

$r = 10$ **[A1]**

$\theta = 36$ **[A1]**

(b) $\dfrac{x+20}{360} \times 2\pi x = \dfrac{16\pi}{5}$ **[M1]**

Simplifying to form a 3-term quadratic equation, e.g. $x^2 + 20x - 576 = 0$ **[M1]**

Valid method to solve the quadratic (e.g. $(x+36)(x-16) = 0$) **[M1]**

$x = 16$ (with -36 rejected or simply 16 as the final answer) **[A1]**

[Total Marks: 9]

6 (a) $\dfrac{72}{360} \times \pi \times r^2 = 85$ **[M1]**

$r^2 = \dfrac{85 \times 5}{\pi}$ or $135.28\ldots$ **[M1]**

11.6 or $11.63\ldots$ **[A1]**

(b) Arc length $= \dfrac{72}{360} \times 2 \times \pi \times 11.63$ or $14.6\ldots$ **[M1]**

Perimeter $= 2 \times 11.63 + 14.61$ **[M1]**

37.9 or 37.86 to 37.87 **[A1]**

[Total Marks: 6]

7 (a) $\dfrac{250}{360} \times 2 \times \pi \times r = 35$ **[M1]**

$r = \dfrac{35 \times 360}{250 \times 2 \times \pi}$ **[M1]**

8.02 or 8.021... **[A1]**

(b) $\dfrac{250}{360} \times \pi \times (\text{their } 8.02)^2$ **[M1]**

OR $\dfrac{1}{2} \times \text{their } 8.02 \times 35$ **[M1]**

140 or 140.3 to 140.4 **[A1]**

[Total Marks: 6]

8 (a) $\dfrac{120}{360} \times \pi \times 9^2$ OR $\dfrac{120}{360} \times \pi \times 5^2$ **[M1]**

$\dfrac{120}{360} \times \pi \times 9^2 - \dfrac{120}{360} \times \pi \times 5^2$ **[M1]**

58.6 $(58.64\ldots)$ OR $\dfrac{56\pi}{3}$ **[A1]**

(b) $\dfrac{120}{360} \times 2 \times \pi \times 9$ OR $\dfrac{120}{360} \times 2 \times \pi \times 5$ **[M1]**

$\dfrac{120}{360} \times 2 \times \pi \times 9 + \dfrac{120}{360} \times 2 \times \pi \times 5 + 4 + 4$ **[M1]**

37.3 $(37.32\ldots)$ OR $\dfrac{28\pi}{3} + 8$ **[A1]**

(c) $\pi r^2 = 58.64\ldots$ (or their area from (a)) **[M1]**

4.32 $(4.320\ldots)$ **[A1]**

[Total Marks: 8]

9 (a) $\frac{110}{360} \times \pi \times 8.5^2$ **[M1]**

69.4 (or $69.35\ldots$) **[A1]**

(b) Area of triangle $OAB = \frac{1}{2} \times 8.5^2 \times \sin(110°)$ $(=33.94\ldots)$ **[M1]**

Area of segment $= 69.35\cdots - 33.94\ldots$ **[M1]**

35.4 (or $35.40\ldots$) **[A1]**

(c) Arc length $AB = \frac{110}{360} \times 2 \times \pi \times 8.5$ $(=16.31\ldots)$ **[M1]**

Chord $AB = \sqrt{8.5^2 + 8.5^2 - 2(8.5)(8.5)\cos(110°)}$ OR $2 \times 8.5\sin(55°)$ **[M1]**

Chord $AB = 13.92\ldots$ **[A1]**

Perimeter $= 16.31\cdots + 13.92\cdots = 30.2$ (or $30.24\ldots$) **[A1]**

[Total Marks: 9]

10 (a) Area of sector $= \frac{110}{360} \times \pi \times 8^2$ $(= 61.43...)$ **[M1]**

Area of triangle $= \frac{1}{2} \times 8 \times 8 \times \sin(110°)$ $(= 30.07...)$ **[M1]**

Area of segment = Area of sector − Area of triangle **[M1]**

31.4 or 31.36... **[A1]**

(b) Arc length $= \frac{110}{360} \times 2 \times \pi \times 8$ $(= 15.35...)$ **[M1]**

Chord $AB = \sqrt{8^2 + 8^2 - 2 \times 8 \times 8 \times \cos(110°)}$ OR $2 \times 8 \times \sin(55°)$ $(= 13.11...)$ **[M1]**

28.5 or 28.46... **[A1]**

[Total Marks: 7]

11 (a) (i) $\frac{x}{360} \times 2\pi r = 6\pi$ OR $\frac{x}{360} \times \pi r^2 = 27\pi$ **[M1]**

Valid method to eliminate x, e.g. dividing the two equations: $\frac{\frac{x}{360}\pi r^2}{\frac{x}{360}2\pi r} = \frac{27\pi}{6\pi}$ leading to $\frac{r}{2} = 4.5$ **[M1]**

9 **[A1]**

(ii) Substitutes their r back into an equation, e.g. $\frac{x}{360} \times 2\pi(9) = 6\pi$ **[M1]**

120 **[A1]**

(b) Perimeter of sector $= \frac{y}{360} \times 2\pi R + 2R$ **[M1]**

Equates to circumference: $\frac{y}{360} \times 2\pi R + 2R = 2\pi R$ **[M1]**

Rearranges to isolate terms with y: $\frac{y}{360} \times 2\pi R = 2\pi R - 2R$ **[M1]**

$\frac{360(\pi - 1)}{\pi}$ OR $360 - \frac{360}{\pi}$ **[A1]**

[Total Marks: 9]

12 (a) Uses cosine rule: $\cos(AOB) = \frac{15^2 + 15^2 - 24^2}{2 \times 15 \times 15}$ or $24^2 = 15^2 + 15^2 - 2(15)(15)\cos(AOB)$ **[M1]**

$\cos(AOB) = -0.28$ **[M1]**

$106.26\ldots$ which rounds to $106.3°$ with no errors seen **[A1]**

(Alternative: $\sin(\frac{AOB}{2}) = \frac{12}{15}$ [M1], $\frac{AOB}{2} = 53.13\ldots°$ [M1], $AOB = 106.3°$ [A1])

(b) (i) Arc length $= \frac{106.3}{360} \times 2 \times \pi \times 15$ $(= 27.8\ldots)$ **[M1]**

Perimeter $= 24 + 27.8\ldots$ **[M1]**

51.8 (or 51.81 to 51.83) **[A1]**

(ii) Area of sector $= \frac{106.3}{360} \times \pi \times 15^2$ $(= 208.6\ldots$ to $208.7\ldots)$ **[M1]**

Area of triangle $= \frac{1}{2} \times 15 \times 15 \times \sin(106.3°)$ $(= 108.0\ldots)$ **[M1]**

Subtracts triangle area from sector area: $208.7 - 108.0$ **[M1]**

101 or 100.6 to 100.8 **[A1]**

[Total Marks: 10]